26 · Smell: the olfactory epithelium

The words of this track, as it says them. Bold brown is the examiner.

Smell. Five marks.

Briefly describe the structure of the olfactory epithelium, and discuss the physiology of olfaction.

Taste's twin, and also her learning outcome, word for word.

The hook: roof, mucus, receptor, cribriform plate, bulb, temporal lobe.

The structure. The olfactory epithelium lies in the roof of the nasal cavity. It sits so high that most of the air you breathe never reaches it, which is why you sniff. It holds the olfactory receptor neurons, which are chemoreceptors with olfactory cilia. Plus supporting cells, and basal cells, which regularly replace the receptor neurons.

Olfaction. Odourants are breathed in, and dissolve in the mucus covering the olfactory epithelium. Then they bind to receptors on the olfactory receptor neurons, which starts a signal. The neurons' axons pass up through the cribriform plate as cranial nerve one, the olfactory nerve, and synapse in the olfactory bulb.

The olfactory bulbs lie under the frontal lobe, and the olfactory tract carries the signal on. The primary olfactory cortex, in the temporal lobe, identifies the smell. The amygdala ties smell to emotion, and the hippocampus ties it to memory. That's why a smell can bring a memory straight back.

Complete loss of smell is anosmia. Partial loss is hyposmia.

The traps. Smell is cranial nerve one, the olfactory nerve. Two is the optic nerve. One of her true or false statements says the second cranial nerve. False. Second: the receptors are the olfactory receptor cells and their cilia, not the basal or supporting cells. Third: it's in the roof of the nasal cavity.

Once more. Smell.

The olfactory epithelium is in the roof of the nasal cavity, which is why you sniff. It has olfactory receptor neurons with cilia, supporting cells, and basal cells that replace the neurons. Odourants dissolve in the mucus and bind the receptors. The axons pass through the cribriform plate as cranial nerve one, and synapse in the olfactory bulb, under the frontal lobe. The primary olfactory cortex in the temporal lobe identifies the smell; the amygdala adds emotion, and the hippocampus memory. Total loss is anosmia; partial loss is hyposmia.